



# SIL Verification Report

Hybrit Pilot Plant

| REV | DATE      | APPROVED | DESCRIPTION OF CHANGE                         |
|-----|-----------|----------|-----------------------------------------------|
|     | 6/5/2020  |          | Document created (First calculation of SIF)   |
| 1   | 6/25/2020 |          | Change from E&H Ex Barrier to PR electronics. |
|     |           |          |                                               |

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## 1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the results, selections, and assumptions made during the Safety Integrity Level verification for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project.

### 1.1 General Project Information

Project Identification: 001  
Project Name: Hybrit Pilot Plant  
Revision: 0  
Company:  
Project Description:

## 2 SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) =>Trip red.gaskompressor (346PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function.

### 2.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300 H2) =>Trip red.gaskompressor (346PU101 SIL2) |
| SIF Tag          | SIF-049                                                                                                       |
| SIF Description: | SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300 H2) =>Trip red.gaskompressor (346PU101 SIL2) |

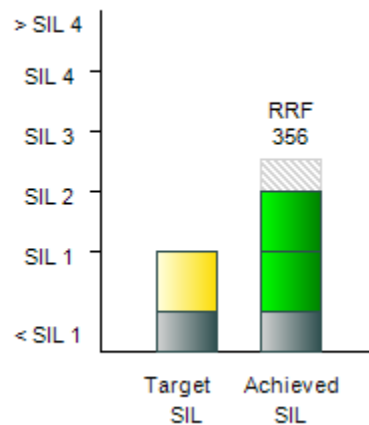
### 2.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF  $\geq 10$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 356.4



### 2.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

exSILentia-VER-002-01-b SILver Detailed-Eng

Mission Time: 15 years  
Startup time: 24 hours

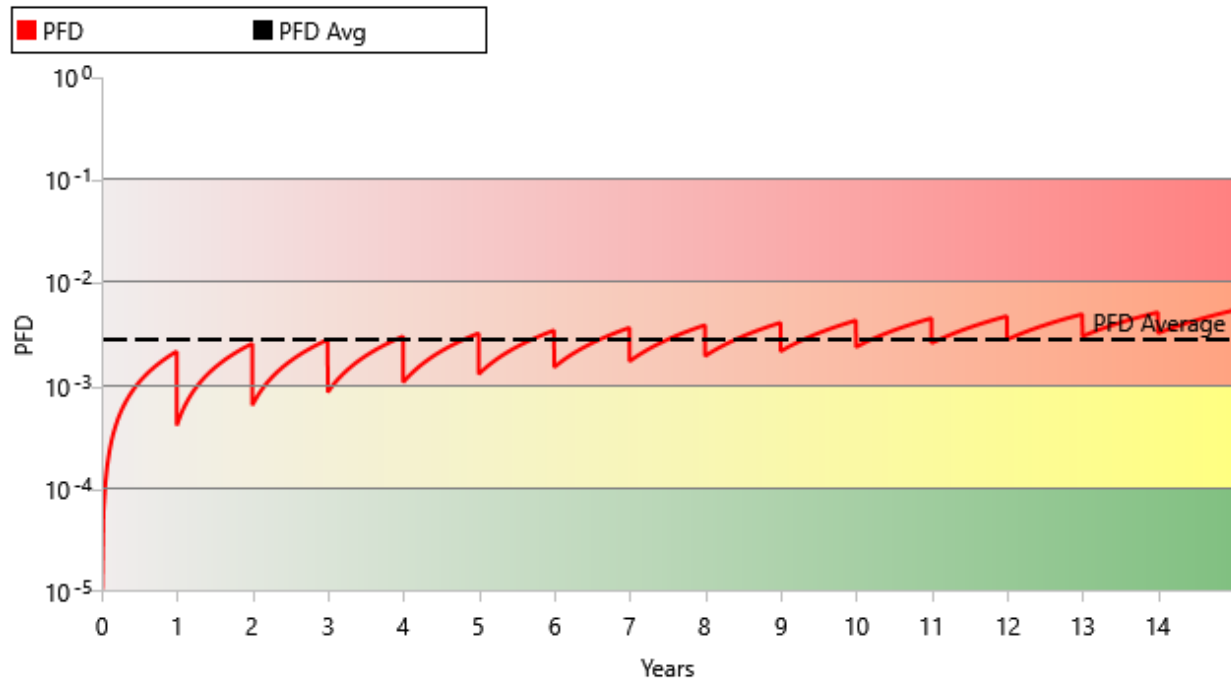
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

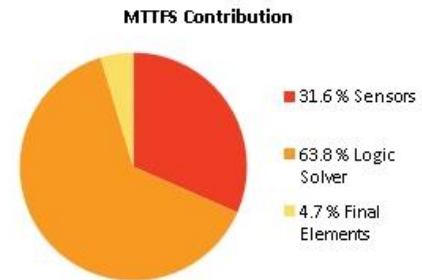
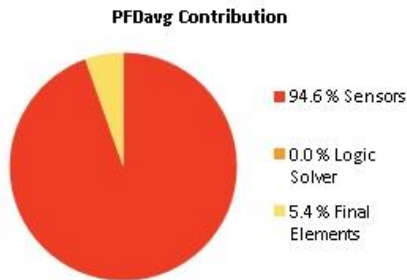
**SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) => Trip red.gaskompressor (346PU101 SIL2)**



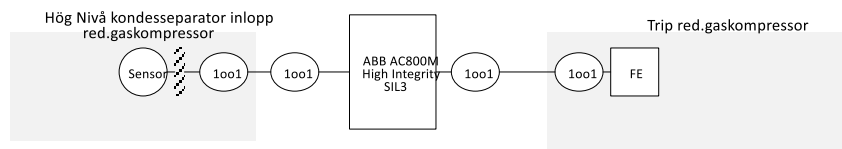
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 1 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 2.81E-03           | 356.4 | 2                  | 2                              | N/A                      | 46.19         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 2.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 2 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.65E-03           | 2                              | N/A                      | 146.26        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 2.3.1.1 Sensor Group 1: Hög Nivå kondesseparator inlopp red.gaskompressor

The information and reliability data underneath describe the Hög Nivå kondesseparator inlopp red.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP



|                                                                                                                                                    |                  |                          |    |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------|----|-----|
| Officine Orobiche TLT                                                                                                                              | -                | -                        | A  | 2   |
| Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off |                  |                          |    |     |
| PR Electronics 9106, 1 active in/1 active out                                                                                                      | 2014.4.03        | -                        | -  | A 2 |
| ABB AC800M High Integrity SIL3 [AnalogIn]                                                                                                          | 2013.1.02        | -                        | -  | B 3 |
| MRT [Hours]:                                                                                                                                       | 24               |                          |    |     |
| Proof Test Interval [Months]:                                                                                                                      | 12               | Proof Test Coverage [%]: | 90 |     |
| Proof Test Execution:                                                                                                                              | Duration 1 hours |                          |    |     |

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög Nivå kondesseparator inlopp red.gaskompressor.

**Table 3 Reliability Data Sensor Group Hög Nivå kondesseparator inlopp red.gaskompressor**

| COMPONENT                                     | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                             |                             |                             |
|-----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
|                                               | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                          | AU                          | NE                          |
| Each Leg                                      |                             |                             |                             |                             |                             |                             |                             |                             |                             |                             |
| Officine Orobiche TLT                         | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 5.71E-07<br><i>6.47E-07</i> | 8.60E-08<br><i>8.60E-08</i> | 7.60E-08<br><i>0.00E+00</i> | 4.80E-08<br><i>4.80E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> |
| PR Electronics 9106, 1 active in/1 active out | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>2.70E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>1.87E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]     | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
| SFF [%]                                       |                             |                             |                             |                             |                             |                             |                             |                             | 76.9                        |                             |
| ROUTE 2 <sub>H</sub> COMPLIANT                |                             |                             |                             |                             |                             |                             |                             |                             | -                           |                             |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 2.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 4 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 4.00E-08           | 3                              | N/A                      | 72.45         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

|                               |                                |                          |           |     |   |         |     |
|-------------------------------|--------------------------------|--------------------------|-----------|-----|---|---------|-----|
| Equipment:                    | ABB AC800M High Integrity SIL3 | SERH VERSION             | 2013.1.02 | PIU | - | SIL Cap | N/A |
| Logic Solver Voting:          |                                | Architecture Type:       | B         |     |   |         |     |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1         |     |   |         |     |
| MRT [Hours]:                  | 24                             |                          |           |     |   |         |     |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 90        |     |   |         |     |
| Proof Test Execution:         | Duration 1 hours               |                          |           |     |   |         |     |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 5 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 2.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-049 SIL1 Hög Nivå kondensseparator inlopp red.gaskompressor (346LI300|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 6 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                  |                       | MTTFS (YEARS) | HFT | SSI |
|--------------------|-----------------------------|-----------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS IEC 61511 | SYSTEMATIC CAPABILITY |               |     |     |
| 1.53E-04           | 2                           | N/A                   | 990.72        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1oo1 |
| $\beta$ -factor [%]:              | N/A  |

#### 2.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

|                      |      |                      |           |
|----------------------|------|----------------------|-----------|
| Voting within group: | 1oo1 | Voting type:         | Identical |
| HFT:                 | 0    | $\beta$ -factor [%]: | N/A       |

Equipment Leg  
(each):

|                                             | SERH VERSION     | 2 <sub>H</sub>           | PIU | AC TYPE | SIL CAP |
|---------------------------------------------|------------------|--------------------------|-----|---------|---------|
| ASC880 R2 STO                               |                  | -                        | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02        | -                        | -   | B       | 3       |
| MRT [Hours]:                                | 24               |                          |     |         |         |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: | 90  |         |         |
| Proof Test Execution:                       | Duration 1 hours |                          |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

**Table 7 Reliability Data Final Element Group Trip red.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

### 3 SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function.

#### 3.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------|
| SIF Name:        | SIL2 Gastryck inlopp red.gaskompressor (346PI304 L2) =>Trip red.gaskompressor (346PU101 SIL2) |
| SIF Tag          | SIF-050                                                                                       |
| SIF Description: | SIL2 Gastryck inlopp red.gaskompressor (346PI304 L2) =>Trip red.gaskompressor (346PU101 SIL2) |

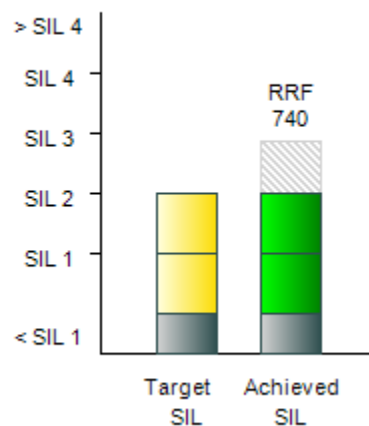
#### 3.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 740.7



#### 3.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

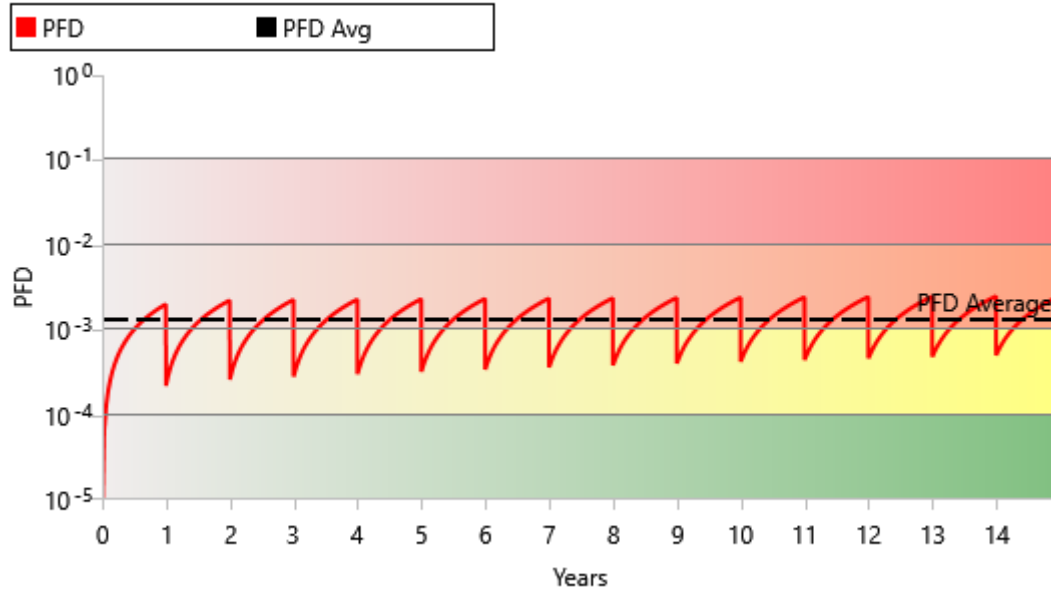
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

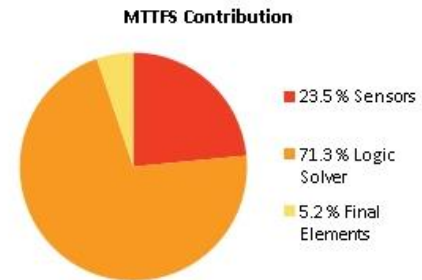
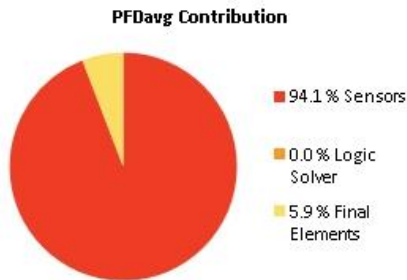
SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) => Trip red.gaskompressor (346PU101 SIL2)



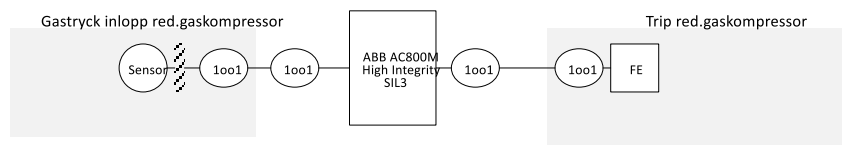
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 8 Functional Safety Performance

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 1.35E-03           | 740.7 | 2                  | 2                              | N/A                      | 51.68         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 3.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-050 SIL2 Gastrick inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 9 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.27E-03           | 2                              | N/A                      | 220.37        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 3.3.1.1 Sensor Group 1: Gastrick inlopp red.gaskompressor

The information and reliability data underneath describe the Gastrick inlopp red.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION      2H      PIU      AC TYPE      SIL CAP

Endress+Hauser Cerabar S PMC71 Standard 2006.2.02 - - B 2  
Exi

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

PR Electronics 9106, 1 active in/1 active out 2014.4.03 - - A 2

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck inlopp red.gaskompressor.

**Table 10 Reliability Data Sensor Group Gastryck inlopp red.gaskompressor**

| COMPONENT                                     | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                                |                             |                             |
|-----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|-----------------------------|
|                                               | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                             | AU                          | NE                          |
| Each Leg                                      |                             |                             |                             |                             |                             |                             |                             |                                |                             |                             |
| Endress+Hauser Cerabar S PMC71 Standard Exi   | 5.21E-08<br><i>0.00E+00</i> | 6.85E-08<br><i>0.00E+00</i> | 2.98E-07<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>3.50E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 8.00E-08<br><i>1.49E-07</i> | 0.00E+00<br><i>0.00E+00</i>    | 1.00E-08<br><i>1.00E-08</i> | 3.82E-07<br><i>3.82E-07</i> |
| PR Electronics 9106, 1 active in/1 active out | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>1.46E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>6.80E-08</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]     | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                               |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        |                             | 71.0                        |
|                                               |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 3.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 11 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 4.00E-08           | 3                              | N/A                      | 72.45         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High

### Integrity SIL3

|                               |                                |                          |     |         |
|-------------------------------|--------------------------------|--------------------------|-----|---------|
| Equipment:                    |                                | SERH VERSION             | PIU | SIL Cap |
|                               | ABB AC800M High Integrity SIL3 | 2013.1.02                | -   | N/A     |
| Logic Solver Voting:          |                                | Architecture Type:       | B   |         |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1   |         |
| MRT [Hours]:                  | 24                             |                          |     |         |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 90  |         |
| Proof Test Execution:         | Duration 1 hours               |                          |     |         |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 12 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 3.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 13 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 7.93E-05           | 2                              | N/A                      | 990.65        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1oo1 |
| $\beta$ -factor [%]:              | N/A  |

#### 3.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

|                      |      |                      |           |
|----------------------|------|----------------------|-----------|
| Voting within group: | 1oo1 | Voting type:         | Identical |
| HFT:                 | 0    | $\beta$ -factor [%]: | N/A       |



Equipment Leg  
(each):

|                                             | SERH VERSION     | 2 <sub>H</sub>           | PIU | AC TYPE | SIL CAP |
|---------------------------------------------|------------------|--------------------------|-----|---------|---------|
| ASC880 R2 STO                               |                  | -                        | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02        | -                        | -   | B       | 3       |
| MRT [Hours]:                                | 24               |                          |     |         |         |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: | 99  |         |         |
| Proof Test Execution:                       | Duration 1 hours |                          |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

**Table 14 Reliability Data Final Element Group Trip red.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 4 SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) =>Trip red.gaskompressor (346PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function.

### 4.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410 H2) =>Trip red.gaskompressor (346PU101 SIL2) |
| SIF Tag          | SIF-051                                                                                                       |
| SIF Description: | SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410 H2) =>Trip red.gaskompressor (346PU101 SIL2) |

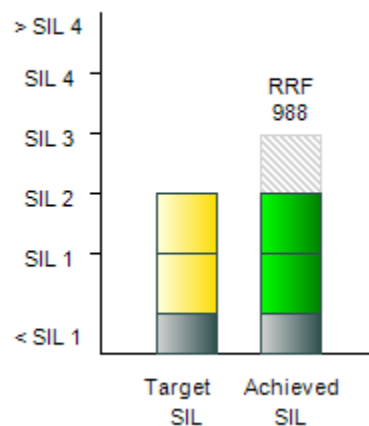
### 4.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 988.2



### 4.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years  
Startup time: 24 hours

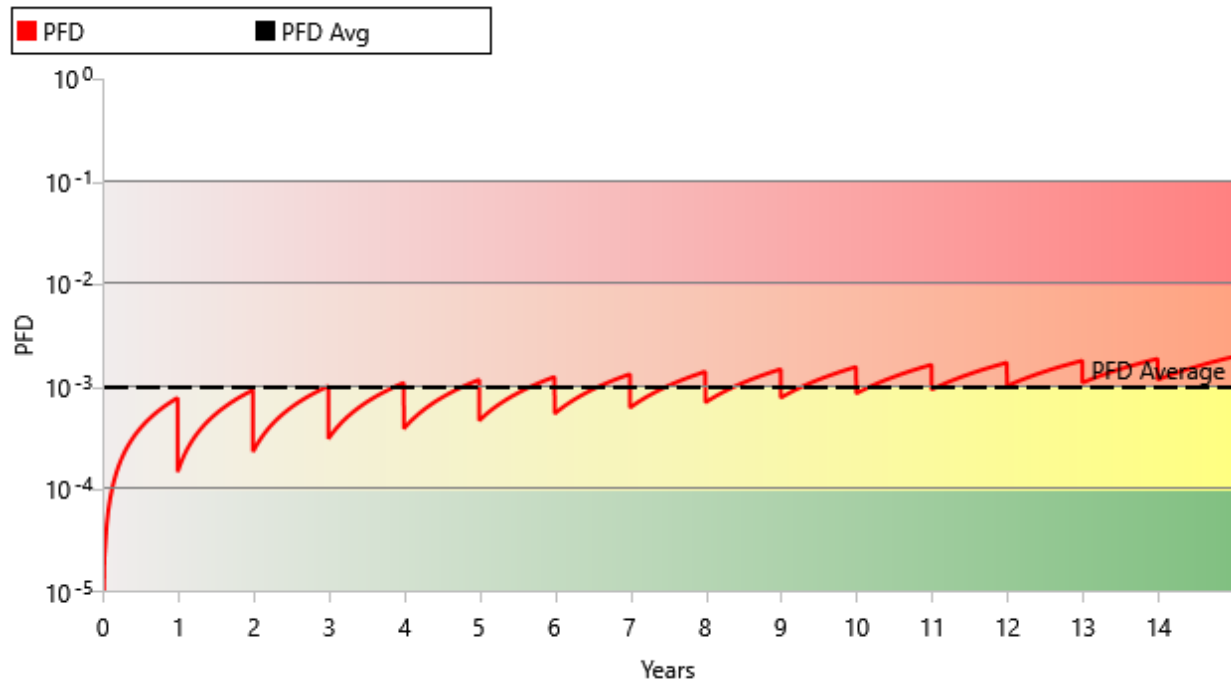
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

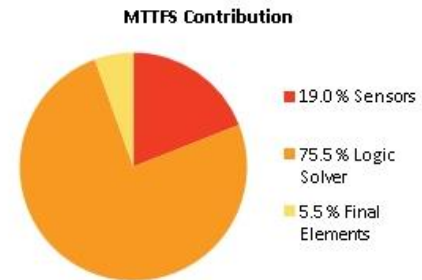
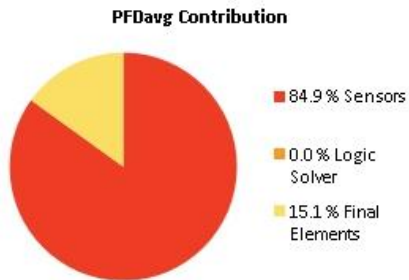
**SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) => Trip red.gaskompressor (346PU101 SIL2)**



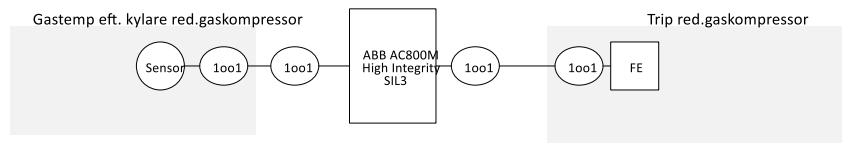
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 15 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 1.01E-03           | 988.2 | 2                  | 2                              | N/A                      | 54.68         |



The SIF Conceptual Design is modeled as shown in the picture below.



#### 4.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 16 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 8.59E-04           | 2                              | N/A                      | 287.37        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

##### 4.3.1.1 Sensor Group 1: Gastemp eft. kylare red.gaskompressor

The information and reliability data underneath describe the Gastemp eft. kylare red.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION      2H      PIU      AC TYPE      SIL CAP

PR electronics PR5435 / PR5437, Single, 4..20mA 2019.4.01 - - B 3

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastemp eft. kylare red.gaskompressor.

**Table 17 Reliability Data Sensor Group Gastemp eft. kylare red.gaskompressor**

| COMPONENT                                       | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                             |                             |                             |
|-------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
|                                                 | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                          | AU                          | NE                          |
| Each Leg                                        |                             |                             |                             |                             |                             |                             |                             |                             |                             |                             |
| PR electronics PR5435 / PR5437, Single, 4..20mA | 4.90E-08<br><i>0.00E+00</i> | 1.70E-08<br><i>0.00E+00</i> | 3.58E-07<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>3.75E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 2.70E-08<br><i>7.60E-08</i> | 1.90E-08<br><i>1.90E-08</i> | 1.10E-08<br><i>1.10E-08</i> | 2.13E-07<br><i>2.13E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]       | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                                 |                             |                             |                             |                             |                             |                             |                             | SFF [%]                     |                             | 84.9                        |
|                                                 |                             |                             |                             |                             |                             |                             |                             | ROUTE 2H COMPLIANT          |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 4.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 18 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                  |                       | MTTFS (YEARS) | HFT | SSI |
|--------------------|-----------------------------|-----------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS IEC 61511 | SYSTEMATIC CAPABILITY |               |     |     |
| 2.01E-08           | 3                           | N/A                   | 72.46         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

|                                |              |     |         |
|--------------------------------|--------------|-----|---------|
| Equipment:                     | SERH VERSION | PIU | SIL Cap |
| ABB AC800M High Integrity SIL3 | 2013.1.02    | -   | N/A     |

Logic Solver Voting: Architecture Type: B  
 HFT: 1  $\beta$ -factor [%]: 1  
 MRT [Hours]: 24  
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99  
 Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 19 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 4.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 20 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                  |                       | MTTFS (YEARS) | HFT | SSI |
|--------------------|-----------------------------|-----------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS IEC 61511 | SYSTEMATIC CAPABILITY |               |     |     |
| 1.53E-04           | 2                           | N/A                   | 990.72        | 0   | 2   |

Number of Final Element group(s): 1  
 Voting between groups: 1001  
 $\beta$ -factor [%]: N/A

#### 4.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical  
 HFT: 0  $\beta$ -factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION 2H PIU AC TYPE SIL CAP

|                                             |                  |                          |    |   |
|---------------------------------------------|------------------|--------------------------|----|---|
| ASC880 R2 STO                               | -                | -                        | A  | 3 |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02        | -                        | B  | 3 |
| MRT [Hours]:                                | 24               |                          |    |   |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: | 90 |   |
| Proof Test Execution:                       | Duration 1 hours |                          |    |   |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

**Table 21 Reliability Data Final Element Group Trip red.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 5 SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function.

### 5.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311 H2) =>Trip red.gaskompressor (346PU101 SIL2) |
| SIF Tag          | SIF-052                                                                                                        |
| SIF Description: | SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311 H2) =>Trip red.gaskompressor (346PU101 SIL2) |

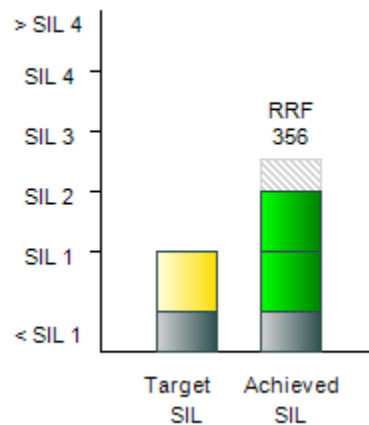
### 5.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF  $\geq 10$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 356.4



### 5.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.



Mission Time: 15 years  
Startup time: 24 hours

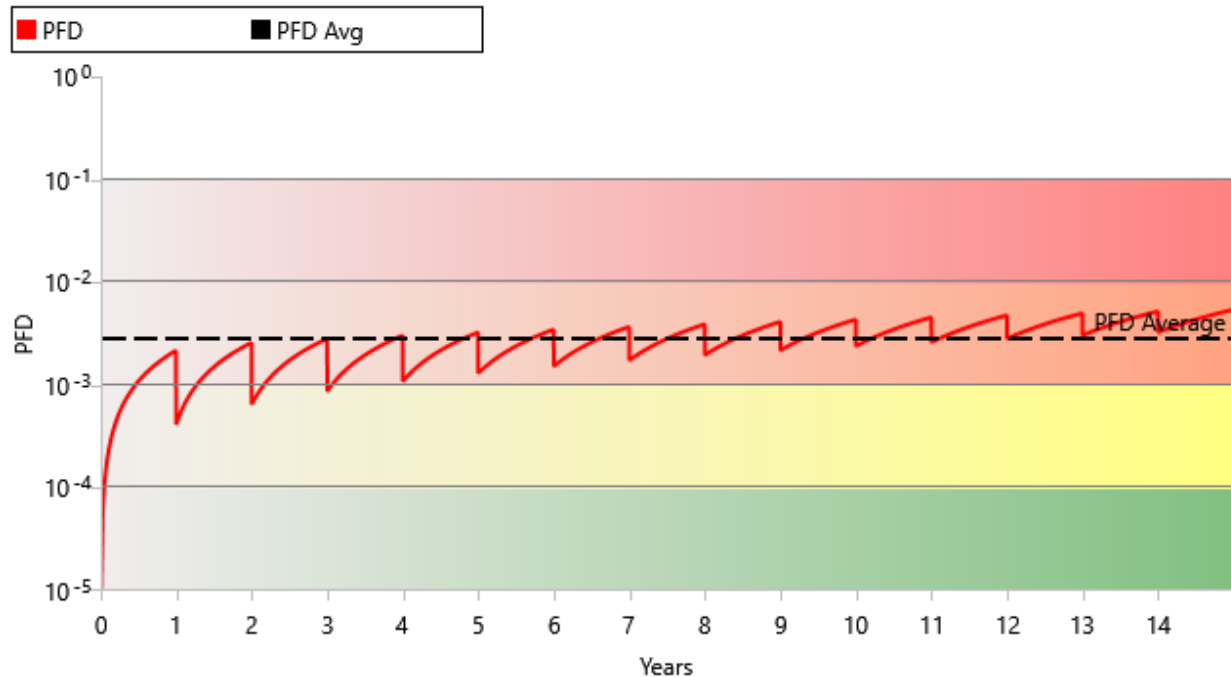
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

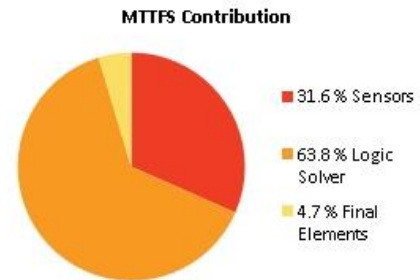
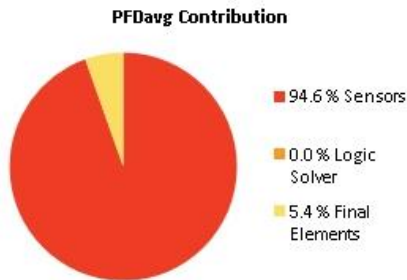
**SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) => Trip red.gaskompressor (346PU101 SIL2)**



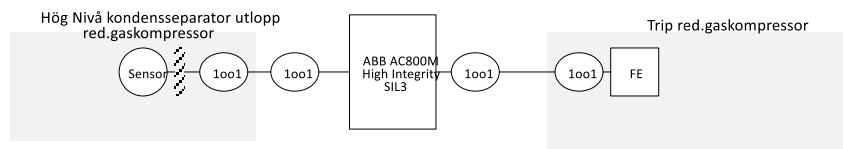
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 22 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 2.81E-03           | 356.4 | 2                  | 2                              | N/A                      | 46.2          |



The SIF Conceptual Design is modeled as shown in the picture below.



### 5.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 23 Sensor Part Functional Safety Performance**

| PFDavg   | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|----------|--------------------------------|--------------------------|---------------|-----|-----|
|          | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.65E-03 | 2                              | N/A                      | 146.26        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1001  
β-factor [%]: N/A

#### 5.3.1.1 Sensor Group 1: Hög Nivå kondensseparator utlopp red.gaskompressor

The information and reliability data underneath describe the Hög Nivå kondensseparator utlopp red.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

|                                                                                                                                                    |                  |                          |    |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------|----|-----|
| Officine Orobiche TLT                                                                                                                              | -                | -                        | A  | 2   |
| Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off |                  |                          |    |     |
| PR Electronics 9106, 1 active in/1 active out                                                                                                      | 2014.4.03        | -                        | -  | A 2 |
| ABB AC800M High Integrity SIL3 [AnalogIn]                                                                                                          | 2013.1.02        | -                        | -  | B 3 |
| MRT [Hours]:                                                                                                                                       | 24               |                          |    |     |
| Proof Test Interval [Months]:                                                                                                                      | 12               | Proof Test Coverage [%]: | 90 |     |
| Proof Test Execution:                                                                                                                              | Duration 1 hours |                          |    |     |

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög Nivå kondensseparator utlopp red.gaskompressor.

**Table 24 Reliability Data Sensor Group Hög Nivå kondensseparator utlopp red.gaskompressor**

| COMPONENT                                     | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                                |                             |                             |
|-----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|-----------------------------|
|                                               | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                             | AU                          | NE                          |
| Each Leg                                      |                             |                             |                             |                             |                             |                             |                             |                                |                             |                             |
| Officine Orobiche TLT                         | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 5.71E-07<br><i>6.47E-07</i> | 8.60E-08<br><i>8.60E-08</i> | 7.60E-08<br><i>0.00E+00</i> | 4.80E-08<br><i>4.80E-08</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> |
| PR Electronics 9106, 1 active in/1 active out | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>2.70E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>1.87E-07</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]     | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                               |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        |                             | 76.9                        |
|                                               |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 5.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 25 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.01E-08           | 3                              | N/A                      | 72.46         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

|                               |                                |                          |           |     |   |         |     |
|-------------------------------|--------------------------------|--------------------------|-----------|-----|---|---------|-----|
| Equipment:                    | ABB AC800M High Integrity SIL3 | SERH VERSION             | 2013.1.02 | PIU | - | SIL Cap | N/A |
| Logic Solver Voting:          |                                | Architecture Type:       | B         |     |   |         |     |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1         |     |   |         |     |
| MRT [Hours]:                  | 24                             |                          |           |     |   |         |     |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 99        |     |   |         |     |
| Proof Test Execution:         | Duration 1 hours               |                          |           |     |   |         |     |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 26 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 5.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 27 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.53E-04           | 2                              | N/A                      | 990.72        | 0   | 2   |

Number of Final Element group(s): 1  
Voting between groups: 1oo1  
 $\beta$ -factor [%]: N/A

#### 5.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical

|                                             |                  |                          |                |     |         |         |
|---------------------------------------------|------------------|--------------------------|----------------|-----|---------|---------|
| HFT:                                        | 0                | $\beta$ -factor [%]:     |                | N/A |         |         |
| Equipment Leg (each):                       |                  | SERH VERSION             | 2 <sub>H</sub> | PIU | AC TYPE | SIL CAP |
| ASC880 R2 STO                               |                  |                          | -              | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02        |                          | -              | -   | B       | 3       |
| MRT [Hours]:                                | 24               |                          |                |     |         |         |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: |                |     | 90      |         |
| Proof Test Execution:                       | Duration 1 hours |                          |                |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

**Table 28 Reliability Data Final Element Group Trip red.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 6 SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function.

### 6.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                              |
|------------------|----------------------------------------------------------------------------------------------|
| SIF Name:        | SIL2 Lågt oljetryck red.gaskompressor (346PI334 L2) =>Trip red.gaskompressor (346PU101 SIL2) |
| SIF Tag          | SIF-053                                                                                      |
| SIF Description: | SIL2 Lågt oljetryck red.gaskompressor (346PI334 L2) =>Trip red.gaskompressor (346PU101 SIL2) |

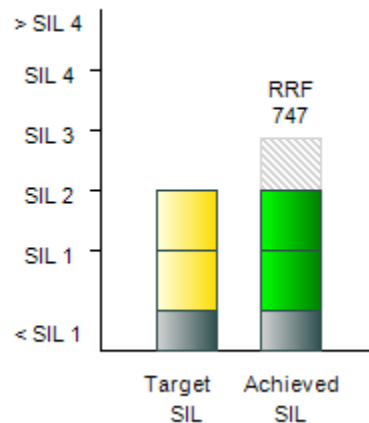
### 6.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 747.9



### 6.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years  
Startup time: 24 hours

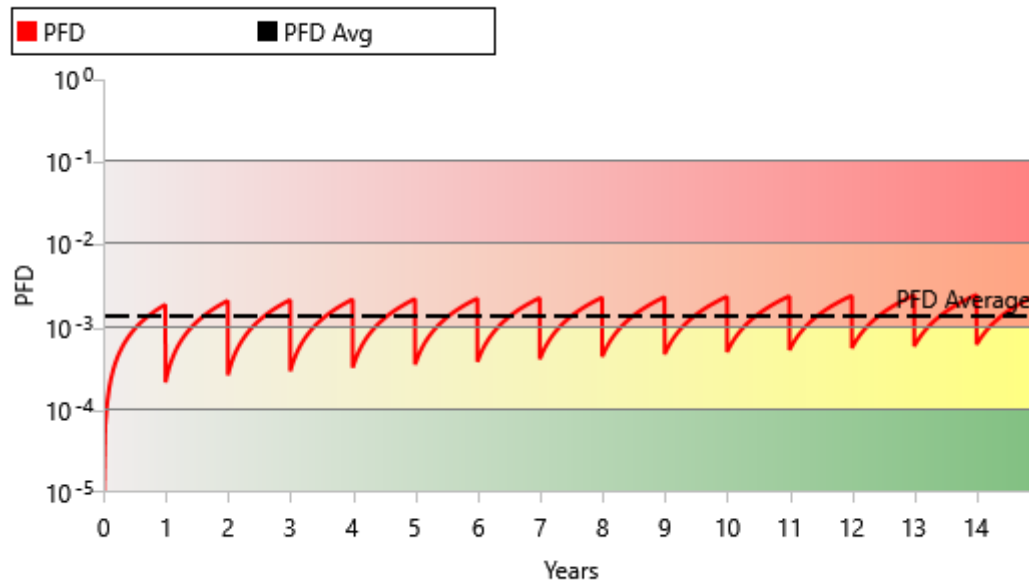
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

**SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) => Trip red.gaskompressor (346PU101 SIL2)**



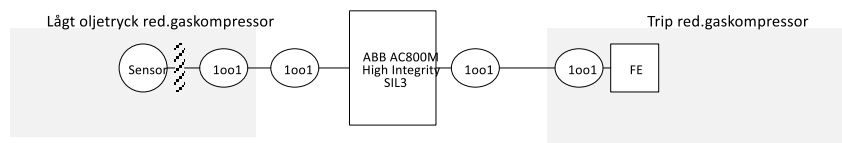
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) => Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 29 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 1.34E-03           | 747.9 | 2                  | 2                              | N/A                      | 51.02         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 6.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 30 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.18E-03           | 2                              | N/A                      | 208.68        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 6.3.1.1 Sensor Group 1: Lågt oljetryck red.gaskompressor

The information and reliability data underneath describe the Lågt oljetryck red.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION      2H      PIU      AC TYPE      SIL CAP



Endress+Hauser Cerabar S PMP71, PMP72  
HT, PMP75 2006.2.02 - - B 2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

PR Electronics 9106, 1 active in/1 active out 2014.4.03 - - A 2

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Lågt oljetryck red.gaskompressor.

**Table 31 Reliability Data Sensor Group Lågt oljetryck red.gaskompressor**

| COMPONENT                                       | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                                |                             |                             |
|-------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|-----------------------------|
|                                                 | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                             | AU                          | NE                          |
| Each Leg                                        |                             |                             |                             |                             |                             |                             |                             |                                |                             |                             |
| Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75 | 5.21E-08<br><i>0.00E+00</i> | 6.85E-08<br><i>0.00E+00</i> | 3.27E-07<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>3.79E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 6.53E-08<br><i>1.34E-07</i> | 0.00E+00<br><i>0.00E+00</i>    | 1.00E-08<br><i>1.00E-08</i> | 3.03E-07<br><i>3.03E-07</i> |
| PR Electronics 9106, 1 active in/1 active out   | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>1.46E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>6.80E-08</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]       | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                                 |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        |                             | 73.4                        |
|                                                 |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 6.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 32 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.01E-08           | 3                              | N/A                      | 72.46         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High

### Integrity SIL3

|                               |                                |                          |     |         |
|-------------------------------|--------------------------------|--------------------------|-----|---------|
| Equipment:                    |                                | SERH VERSION             | PIU | SIL Cap |
|                               | ABB AC800M High Integrity SIL3 | 2013.1.02                | -   | N/A     |
| Logic Solver Voting:          |                                | Architecture Type:       | B   |         |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1   |         |
| MRT [Hours]:                  | 24                             |                          |     |         |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 99  |         |
| Proof Test Execution:         | Duration 1 hours               |                          |     |         |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 33 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
| SFF [%]                                              |                      |          |          |          |          |          | 100.0    |

### 6.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 34 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                  |                       | MTTFS (YEARS) | HFT | SSI |
|--------------------|-----------------------------|-----------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS IEC 61511 | SYSTEMATIC CAPABILITY |               |     |     |
| 1.53E-04           | 2                           | N/A                   | 990.72        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1oo1 |
| $\beta$ -factor [%]:              | N/A  |

#### 6.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0       $\beta$ -factor [%]: N/A

|                                             |                     |                             |            |                |                |
|---------------------------------------------|---------------------|-----------------------------|------------|----------------|----------------|
| Equipment Leg<br>(each):                    | <b>SERH VERSION</b> | <b>2<sub>H</sub></b>        | <b>PIU</b> | <b>AC TYPE</b> | <b>SIL CAP</b> |
| ASC880 R2 STO                               |                     | -                           | -          | A              | 3              |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02           | -                           | -          | B              | 3              |
| MRT [Hours]: 24                             |                     |                             |            |                |                |
| Proof Test Interval [Months]: 12            |                     | Proof Test Coverage [%]: 90 |            |                |                |
| Proof Test Execution:                       | Duration 1 hours    |                             |            |                |                |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

**Table 35 Reliability Data Final Element Group Trip red.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 7 SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function.

### 7.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300 H2) =>Trip proc.gaskompressor (354PU101 SIL2) |
| SIF Tag          | SIF-054                                                                                                          |
| SIF Description: | SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300 H2) =>Trip proc.gaskompressor (354PU101 SIL2) |

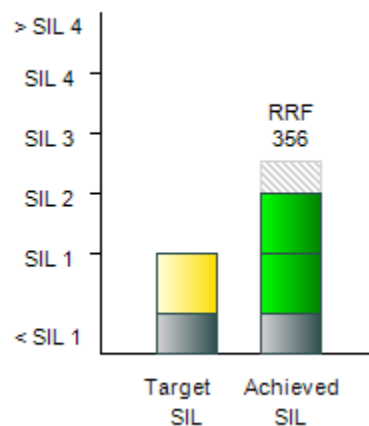
### 7.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF  $\geq 10$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 356.4



### 7.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years  
Startup time: 24 hours

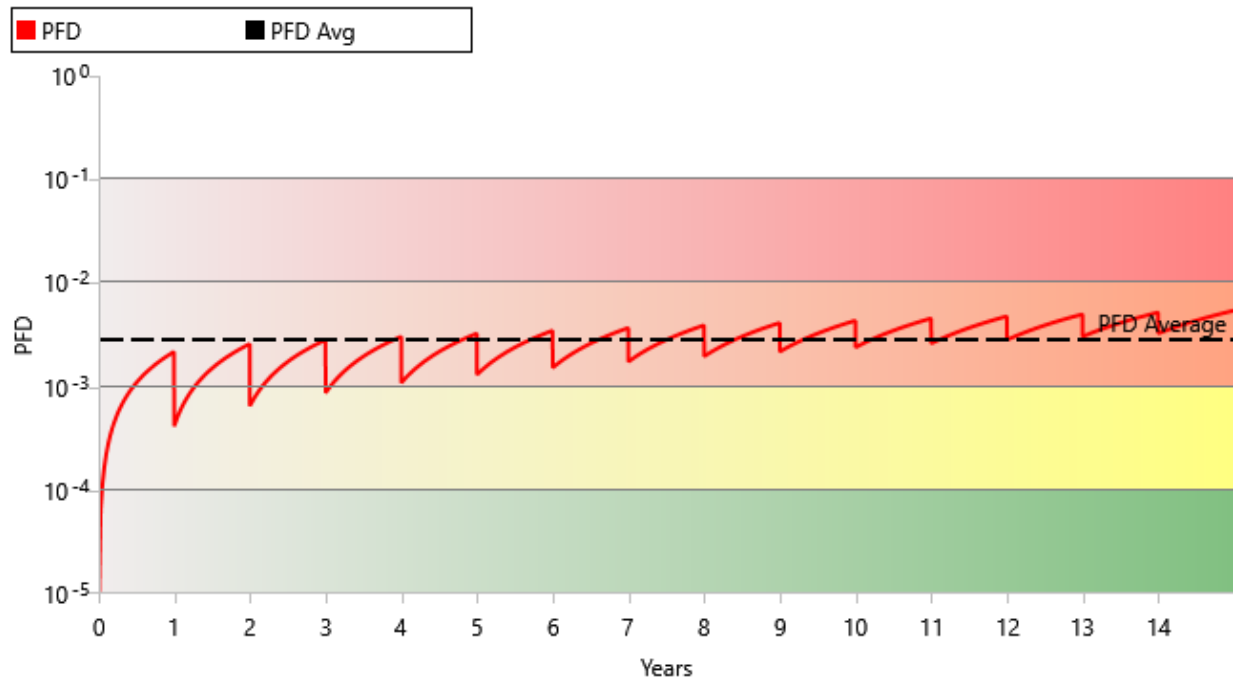
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

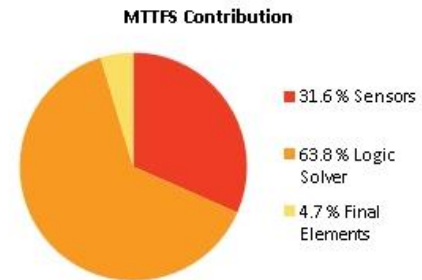
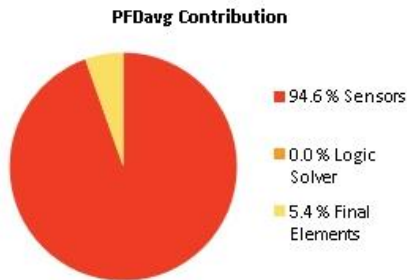
**SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) => Trip proc.gaskompressor (354PU101 SIL2)**



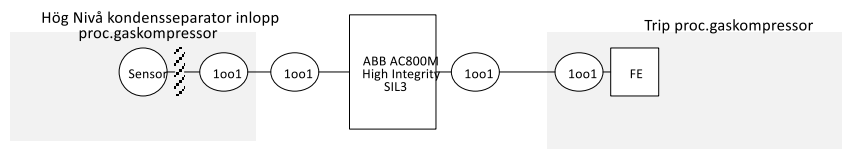
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) => Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 36 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 2.81E-03           | 356.4 | 2                  | 2                              | N/A                      | 46.2          |



The SIF Conceptual Design is modeled as shown in the picture below.



### 7.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 37 Sensor Part Functional Safety Performance**

| PFDavg   | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|----------|--------------------------------|--------------------------|---------------|-----|-----|
|          | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.65E-03 | 2                              | N/A                      | 146.26        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1001  
β-factor [%]: N/A

#### 7.3.1.1 Sensor Group 1: Hög Nivå kondensseparator inlopp proc.gaskompressor

The information and reliability data underneath describe the Hög Nivå kondensseparator inlopp proc.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001      Voting Identical  
type:  
HFT: 0      β-factor N/A  
[%]:

Equipment Leg  
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

|                                                                                                                                                    |                  |                          |    |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------|----|-----|
| Officine Orobiche TLT                                                                                                                              | -                | -                        | A  | 2   |
| Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off |                  |                          |    |     |
| PR Electronics 9106, 1 active in/1 active out                                                                                                      | 2014.4.03        | -                        | -  | A 2 |
| ABB AC800M High Integrity SIL3 [AnalogIn]                                                                                                          | 2013.1.02        | -                        | -  | B 3 |
| MRT [Hours]:                                                                                                                                       | 24               |                          |    |     |
| Proof Test Interval [Months]:                                                                                                                      | 12               | Proof Test Coverage [%]: | 90 |     |
| Proof Test Execution:                                                                                                                              | Duration 1 hours |                          |    |     |

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög Nivå kondensseparator inlopp proc.gaskompressor.

**Table 38 Reliability Data Sensor Group Hög Nivå kondensseparator inlopp proc.gaskompressor**

| COMPONENT                                     | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                             |                                |                             |
|-----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|
|                                               | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                          | AU                             | NE                          |
| Each Leg                                      |                             |                             |                             |                             |                             |                             |                             |                             |                                |                             |
| Officine Orobiche TLT                         | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 5.71E-07<br><i>6.47E-07</i> | 8.60E-08<br><i>8.60E-08</i> | 7.60E-08<br><i>0.00E+00</i> | 4.80E-08<br><i>4.80E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> |
| PR Electronics 9106, 1 active in/1 active out | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>2.70E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>1.87E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i>    | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]     | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i>    | 1.09E-08<br><i>1.09E-08</i> |
|                                               |                             |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        | 76.9                        |
|                                               |                             |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 7.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 39 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.01E-08           | 3                              | N/A                      | 72.46         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

|                               |                                |                          |           |     |   |         |     |
|-------------------------------|--------------------------------|--------------------------|-----------|-----|---|---------|-----|
| Equipment:                    | ABB AC800M High Integrity SIL3 | SERH VERSION             | 2013.1.02 | PIU | - | SIL Cap | N/A |
| Logic Solver Voting:          |                                | Architecture Type:       | B         |     |   |         |     |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1         |     |   |         |     |
| MRT [Hours]:                  | 24                             |                          |           |     |   |         |     |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 99        |     |   |         |     |
| Proof Test Execution:         | Duration 1 hours               |                          |           |     |   |         |     |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 40 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 7.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 41 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.53E-04           | 2                              | N/A                      | 990.72        | 0   | 2   |

Number of Final Element group(s): 1  
Voting between groups: 1oo1  
 $\beta$ -factor [%]: N/A

#### 7.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical



|                                             |                  |                          |                |     |         |         |
|---------------------------------------------|------------------|--------------------------|----------------|-----|---------|---------|
| HFT:                                        | 0                | $\beta$ -factor [%]:     |                | N/A |         |         |
| Equipment Leg (each):                       |                  | SERH VERSION             | 2 <sub>H</sub> | PIU | AC TYPE | SIL CAP |
| ASC880 R2 STO                               |                  |                          | -              | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] |                  | 2013.1.02                | -              | -   | B       | 3       |
| MRT [Hours]:                                | 24               |                          |                |     |         |         |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: |                |     | 90      |         |
| Proof Test Execution:                       | Duration 1 hours |                          |                |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

**Table 42 Reliability Data Final Element Group Trip proc.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 8 SIF-055 SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-055 SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function.

### 8.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL2 Gastryck inlopp proc.gaskompressor (354PI304 L2) =>Trip proc.gaskompressor (354PU101 SIL2) |
| SIF Tag          | SIF-055                                                                                         |
| SIF Description: | SIL2 Gastryck inlopp proc.gaskompressor (354PI304 L2) =>Trip proc.gaskompressor (354PU101 SIL2) |

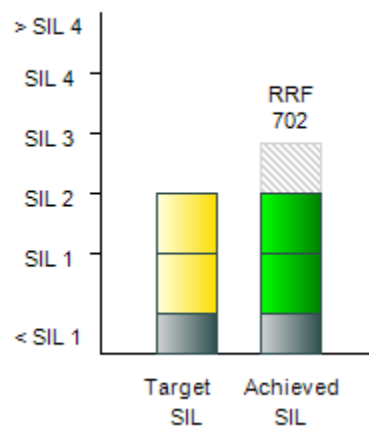
### 8.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 702.6



### 8.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-055 SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

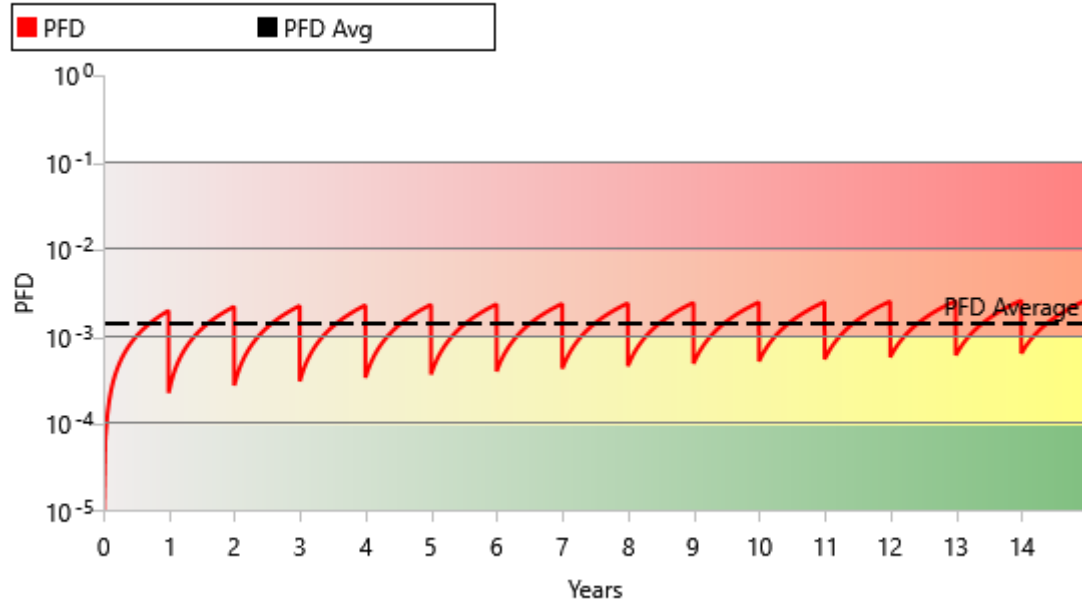
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

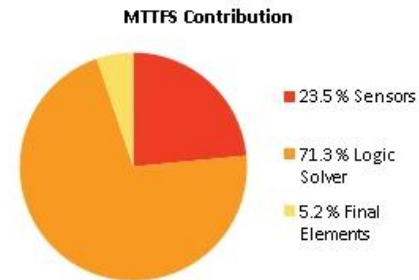
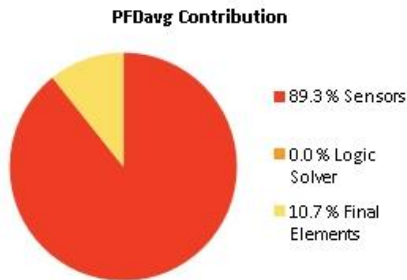
SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) => Trip proc.gaskompressor (354PU101 SIL2)



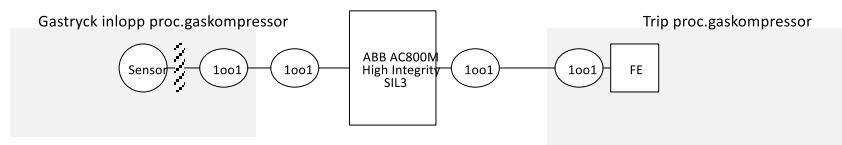
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-055 SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) => Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 43 Functional Safety Performance

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 1.42E-03           | 702.6 | 2                  | 2                              | N/A                      | 51.68         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 8.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-055 SIL2 Gastrick inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 44 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.27E-03           | 2                              | N/A                      | 220.37        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 8.3.1.1 Sensor Group 1: Gastrick inlopp proc.gaskompressor

The information and reliability data underneath describe the Gastrick inlopp proc.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION      2H      PIU      AC TYPE      SIL CAP

Endress+Hauser Cerabar S PMC71 Standard 2006.2.02 - - B 2  
Exi

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

PR Electronics 9106, 1 active in/1 active out 2014.4.03 - - A 2

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastrick inlopp proc.gaskompressor.

**Table 45 Reliability Data Sensor Group Gastrick inlopp proc.gaskompressor**

| COMPONENT                                     | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                                |                             |                             |
|-----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|-----------------------------|
|                                               | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                             | AU                          | NE                          |
| Each Leg                                      |                             |                             |                             |                             |                             |                             |                             |                                |                             |                             |
| Endress+Hauser Cerabar S PMC71 Standard Exi   | 5.21E-08<br><i>0.00E+00</i> | 6.85E-08<br><i>0.00E+00</i> | 2.98E-07<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>3.50E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 8.00E-08<br><i>1.49E-07</i> | 0.00E+00<br><i>0.00E+00</i>    | 1.00E-08<br><i>1.00E-08</i> | 3.82E-07<br><i>3.82E-07</i> |
| PR Electronics 9106, 1 active in/1 active out | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>1.46E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>6.80E-08</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]     | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                               |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        |                             | 71.0                        |
|                                               |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 8.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-055 SIL2 Gastrick inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 46 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 4.00E-08           | 3                              | N/A                      | 72.45         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High

### Integrity SIL3

|                               |                                |                          |     |         |
|-------------------------------|--------------------------------|--------------------------|-----|---------|
| Equipment:                    |                                | SERH VERSION             | PIU | SIL Cap |
|                               | ABB AC800M High Integrity SIL3 | 2013.1.02                | -   | N/A     |
| Logic Solver Voting:          |                                | Architecture Type:       | B   |         |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1   |         |
| MRT [Hours]:                  | 24                             |                          |     |         |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 90  |         |
| Proof Test Execution:         | Duration 1 hours               |                          |     |         |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 47 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 8.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-055 SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 48 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.53E-04           | 2                              | N/A                      | 990.72        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1oo1 |
| $\beta$ -factor [%]:              | N/A  |

#### 8.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

|                      |      |                      |           |
|----------------------|------|----------------------|-----------|
| Voting within group: | 1oo1 | Voting type:         | Identical |
| HFT:                 | 0    | $\beta$ -factor [%]: | N/A       |

Equipment Leg  
(each):

|                                             | SERH VERSION     | 2 <sub>H</sub>           | PIU | AC TYPE | SIL CAP |
|---------------------------------------------|------------------|--------------------------|-----|---------|---------|
| ASC880 R2 STO                               |                  | -                        | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02        | -                        | -   | B       | 3       |
| MRT [Hours]:                                | 24               |                          |     |         |         |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: | 90  |         |         |
| Proof Test Execution:                       | Duration 1 hours |                          |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

**Table 49 Reliability Data Final Element Group Trip proc.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 9 SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function.

### 9.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311 H2) =>Trip proc.gaskompressor (354PU101 SIL2) |
| SIF Tag          | SIF-056                                                                                                          |
| SIF Description: | SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311 H2) =>Trip proc.gaskompressor (354PU101 SIL2) |

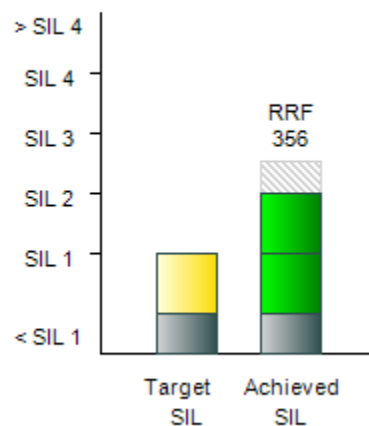
### 9.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF  $\geq 10$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 356.4



### 9.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.



Mission Time: 15 years  
Startup time: 24 hours

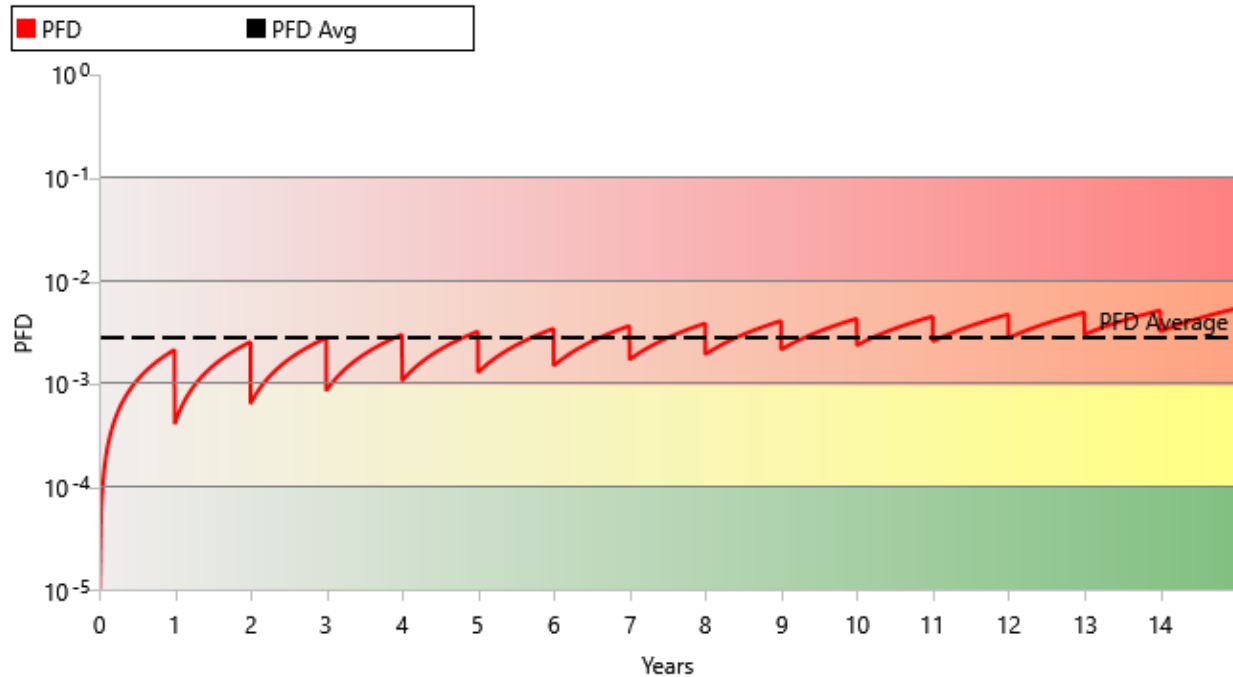
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

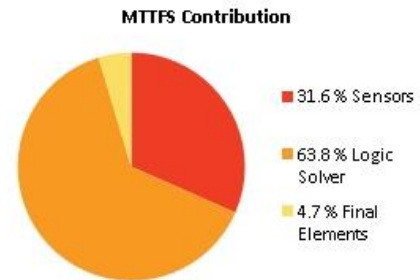
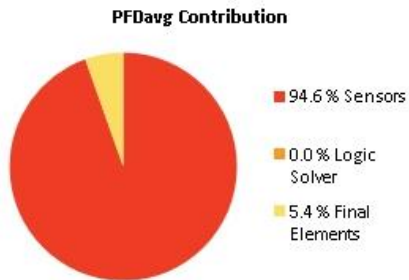
**SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) => Trip proc.gaskompressor (354PU101 SIL2)**



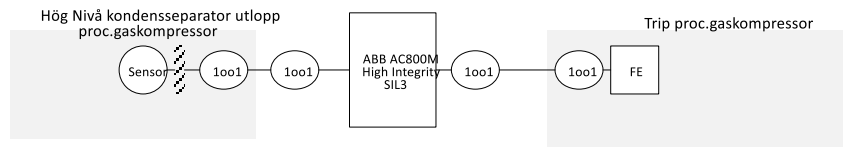
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) => Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 50 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 2.81E-03           | 356.4 | 2                  | 2                              | N/A                      | 46.2          |



The SIF Conceptual Design is modeled as shown in the picture below.



### 9.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table S1 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.65E-03           | 2                              | N/A                      | 146.26        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1001  
β-factor [%]: N/A

#### 9.3.1.1 Sensor Group 1: Hög Nivå kondensseparator utlopp proc.gaskompressor

The information and reliability data underneath describe the Hög Nivå kondensseparator utlopp proc.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001      Voting Identical  
type:  
HFT: 0      β-factor N/A  
[%]:

Equipment Leg  
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

|                                                                                                                                                    |                  |                          |    |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------|----|-----|
| Officine Orobiche TLT                                                                                                                              | -                | -                        | A  | 2   |
| Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off |                  |                          |    |     |
| PR Electronics 9106, 1 active in/1 active out                                                                                                      | 2014.4.03        | -                        | -  | A 2 |
| ABB AC800M High Integrity SIL3 [AnalogIn]                                                                                                          | 2013.1.02        | -                        | -  | B 3 |
| MRT [Hours]:                                                                                                                                       | 24               |                          |    |     |
| Proof Test Interval [Months]:                                                                                                                      | 12               | Proof Test Coverage [%]: | 90 |     |
| Proof Test Execution:                                                                                                                              | Duration 1 hours |                          |    |     |

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög Nivå kondensseparator utlopp proc.gaskompressor.

**Table 52 Reliability Data Sensor Group Hög Nivå kondensseparator utlopp proc.gaskompressor**

| COMPONENT                                     | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                             |                             |                             |
|-----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
|                                               | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                          | AU                          | NE                          |
| Each Leg                                      |                             |                             |                             |                             |                             |                             |                             |                             |                             |                             |
| Officine Orobiche TLT                         | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 5.71E-07<br><i>6.47E-07</i> | 8.60E-08<br><i>8.60E-08</i> | 7.60E-08<br><i>0.00E+00</i> | 4.80E-08<br><i>4.80E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> |
| PR Electronics 9106, 1 active in/1 active out | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>2.70E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>1.87E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]     | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                               |                             |                             |                             |                             |                             |                             |                             |                             | SFF [%]                     | 76.9                        |
|                                               |                             |                             |                             |                             |                             |                             |                             |                             | ROUTE 2h COMPLIANT          | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 9.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 53 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                  |                       | MTTFS (YEARS) | HFT | SSI |
|--------------------|-----------------------------|-----------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS IEC 61511 | SYSTEMATIC CAPABILITY |               |     |     |
| 2.01E-08           | 3                           | N/A                   | 72.46         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

|                               |                                |                          |           |     |   |         |     |
|-------------------------------|--------------------------------|--------------------------|-----------|-----|---|---------|-----|
| Equipment:                    | ABB AC800M High Integrity SIL3 | SERH VERSION             | 2013.1.02 | PIU | - | SIL Cap | N/A |
| Logic Solver Voting:          |                                | Architecture Type:       | B         |     |   |         |     |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1         |     |   |         |     |
| MRT [Hours]:                  | 24                             |                          |           |     |   |         |     |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 99        |     |   |         |     |
| Proof Test Execution:         | Duration 1 hours               |                          |           |     |   |         |     |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 54 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 9.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 55 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.53E-04           | 2                              | N/A                      | 990.72        | 0   | 2   |

Number of Final Element group(s): 1  
Voting between groups: 1oo1  
 $\beta$ -factor [%]: N/A

#### 9.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical

|                                             |                  |                          |                |     |         |         |
|---------------------------------------------|------------------|--------------------------|----------------|-----|---------|---------|
| HFT:                                        | 0                | $\beta$ -factor [%]:     |                | N/A |         |         |
| Equipment Leg<br>(each):                    |                  | SERH VERSION             | 2 <sub>H</sub> | PIU | AC TYPE | SIL CAP |
| ASC880 R2 STO                               |                  |                          | -              | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] |                  | 2013.1.02                | -              | -   | B       | 3       |
| MRT [Hours]:                                | 24               |                          |                |     |         |         |
| Proof Test Interval [Months]:               | 12               | Proof Test Coverage [%]: |                |     | 90      |         |
| Proof Test Execution:                       | Duration 1 hours |                          |                |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

**Table 56 Reliability Data Final Element Group Trip proc.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 10 SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function.

### 10.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                |
|------------------|------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL2 Lågt oljetryck proc.gaskompressor (354PI354 L2) =>Trip proc.gaskompressor (354PU101 SIL2) |
| SIF Tag          | SIF-057                                                                                        |
| SIF Description: | SIL2 Lågt oljetryck proc.gaskompressor (354PI354 L2) =>Trip proc.gaskompressor (354PU101 SIL2) |

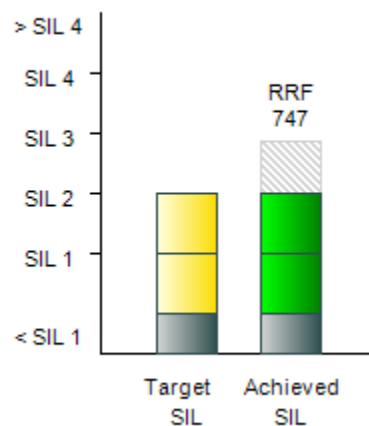
### 10.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 747.9



### 10.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

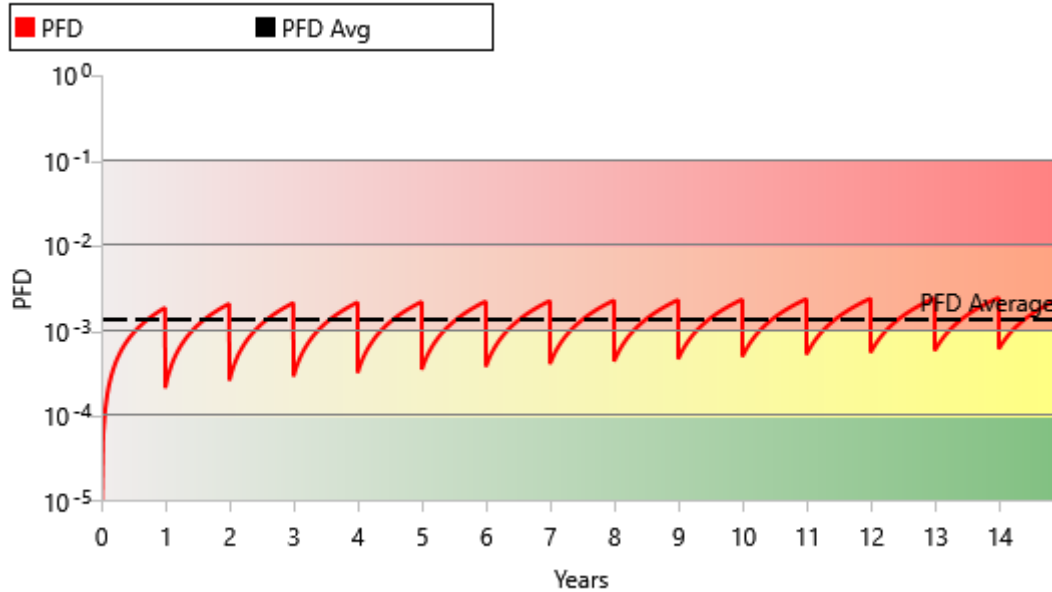
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

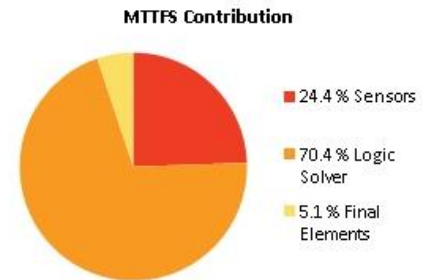
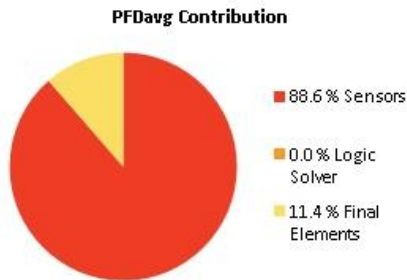
SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) => Trip proc.gaskompressor (354PU101 SIL2)



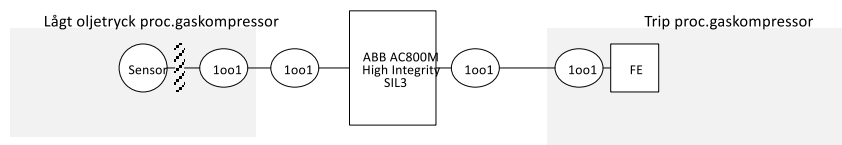
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) => Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 57 Functional Safety Performance

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 1.34E-03           | 747.9 | 2                  | 2                              | N/A                      | 51.01         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 10.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 58 Sensor Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.18E-03           | 2                              | N/A                      | 208.68        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 10.3.1.1 Sensor Group 1: Lågt oljetryck proc.gaskompressor

The information and reliability data underneath describe the Lågt oljetryck proc.gaskompressor sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP



Endress+Hauser Cerabar S PMP71, PMP72  
HT, PMP75 2006.2.02 - - B 2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

PR Electronics 9106, 1 active in/1 active out 2014.4.03 - - A 2

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Lågt oljetryck proc.gaskompressor.

**Table 59 Reliability Data Sensor Group Lågt oljetryck proc.gaskompressor**

| COMPONENT                                       | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                                |                             |                             |
|-------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|-----------------------------|
|                                                 | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                             | AU                          | NE                          |
| Each Leg                                        |                             |                             |                             |                             |                             |                             |                             |                                |                             |                             |
| Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75 | 5.21E-08<br><i>0.00E+00</i> | 6.85E-08<br><i>0.00E+00</i> | 3.27E-07<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>3.79E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 6.53E-08<br><i>1.34E-07</i> | 0.00E+00<br><i>0.00E+00</i>    | 1.00E-08<br><i>1.00E-08</i> | 3.03E-07<br><i>3.03E-07</i> |
| PR Electronics 9106, 1 active in/1 active out   | 1.46E-07<br><i>0.00E+00</i> | 2.70E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>1.46E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 4.10E-08<br><i>6.80E-08</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.77E-07<br><i>1.77E-07</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn]       | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                                 |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        |                             | 73.4                        |
|                                                 |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 10.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 60 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 4.00E-08           | 3                              | N/A                      | 72.45         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High

### Integrity SIL3

|                               |                                |                          |     |         |
|-------------------------------|--------------------------------|--------------------------|-----|---------|
| Equipment:                    |                                | SERH VERSION             | PIU | SIL Cap |
|                               | ABB AC800M High Integrity SIL3 | 2013.1.02                | -   | N/A     |
| Logic Solver Voting:          |                                | Architecture Type:       | B   |         |
| HFT:                          | 1                              | $\beta$ -factor [%]:     | 1   |         |
| MRT [Hours]:                  | 24                             |                          |     |         |
| Proof Test Interval [Months]: | 12                             | Proof Test Coverage [%]: | 90  |         |
| Proof Test Execution:         | Duration 1 hours               |                          |     |         |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 61 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 10.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2) Safety Instrumented Function is quantified as follows.

**Table 62 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                  |                       | MTTFS (YEARS) | HFT | SSI |
|--------------------|-----------------------------|-----------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS IEC 61511 | SYSTEMATIC CAPABILITY |               |     |     |
| 1.53E-04           | 2                           | N/A                   | 990.72        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1oo1 |
| $\beta$ -factor [%]:              | N/A  |

#### 10.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0       $\beta$ -factor [%]: N/A

| Equipment Leg<br>(each):                    | SERH VERSION     | 2 <sub>H</sub>              | PIU | AC TYPE | SIL CAP |
|---------------------------------------------|------------------|-----------------------------|-----|---------|---------|
| ASC880 R2 STO                               |                  | -                           | -   | A       | 3       |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02        | -                           | -   | B       | 3       |
| MRT [Hours]: 24                             |                  |                             |     |         |         |
| Proof Test Interval [Months]: 12            |                  | Proof Test Coverage [%]: 90 |     |         |         |
| Proof Test Execution:                       | Duration 1 hours |                             |     |         |         |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

**Table 63 Reliability Data Final Element Group Trip proc.gaskompressor**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                                |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                             | NE       |
| Each Leg                                    |                      |          |    |          |    |                                |          |
| ASC880 R2 STO                               |                      | 1.08E-08 |    | 1.35E-08 |    |                                |          |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                                | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]                        | 89.5     |
|                                             |                      |          |    |          |    | ROUTE 2 <sub>H</sub> COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 11 SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) Safety Instrumented Function.

### 11.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| SIF Name:        | SIL2 Hög nivå i ångpanna 271AG001 (271LZ316 H2) =><br>Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) |
| SIF Tag          | SIF-076                                                                                                              |
| SIF Description: | SIL2 Hög nivå i ångpanna 271AG001 (271LZ316 H2) =><br>Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) |

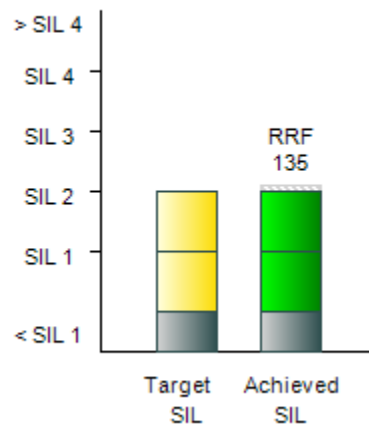
### 11.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 135



### 11.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years  
Startup time: 24 hours

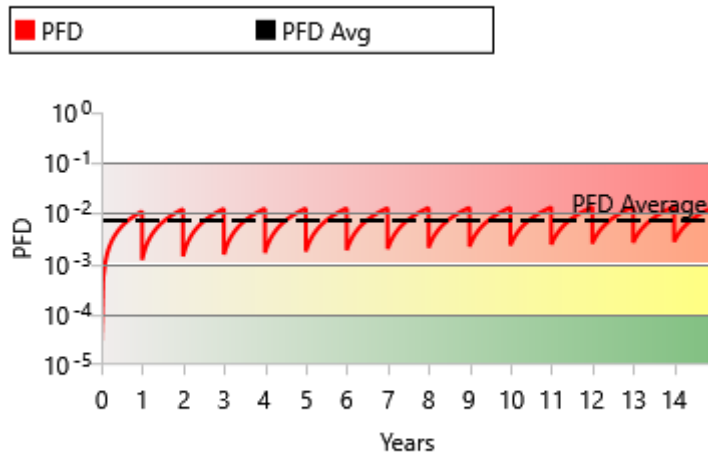
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>  
Close Avstängningsventil 2 överhettad ånga till PGH (271SV556)



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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>

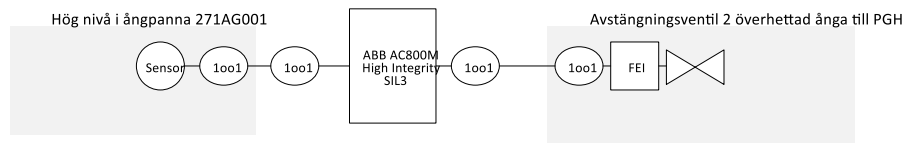
Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 64 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-----|--------------------|--------------------------------|--------------------------|---------------|
|                    |     | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 7.40E-03           | 135 | 2                  | 2                              | N/A                      | 64.74         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 11.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) Safety Instrumented Function is quantified as follows.

**Table 65 Sensor Part Functional Safety Performance**

| PFDavg   | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|----------|--------------------------------|--------------------------|---------------|-----|-----|
|          | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 1.58E-04 | 2                              | N/A                      | 3060.49       | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 11.3.1.1 Sensor Group 1: Hög nivå i ångpanna 271AG001

The information and reliability data underneath describe the Hög nivå i ångpanna 271AG001 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical  
HFT: 0 β-factor [%]: N/A

Equipment Leg  
(each):

Gestra NRS 1-50

SERH VERSION

2<sub>H</sub>

PIU

AC TYPE

SIL CAP

-

-

B

-

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

MRT [Hours]: 24

Proof Test Interval [Months]:

12

Proof Test Coverage [%]:

99

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög nivå i ångpanna 271AG001.

**Table 66 Reliability Data Sensor Group Hög nivå i ångpanna 271AG001**

| COMPONENT       | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                                |                             |                             |
|-----------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------|-----------------------------|-----------------------------|
|                 | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                             | AU                          | NE                          |
| Each Leg        |                             |                             |                             |                             |                             |                             |                             |                                |                             |                             |
| Gestra NRS 1-50 | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 3.73E-08<br><i>3.73E-08</i> | 0.00E+00<br><i>0.00E+00</i> | 2.69E-08<br><i>2.69E-08</i> | 0.00E+00<br><i>0.00E+00</i>    | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> |
|                 |                             |                             |                             |                             |                             |                             |                             | SFF [%]                        |                             | 58.1                        |
|                 |                             |                             |                             |                             |                             |                             |                             | ROUTE 2 <sub>H</sub> COMPLIANT |                             | -                           |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 11.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) Safety Instrumented Function is quantified as follows.

**Table 67 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 6.53E-09           | 3                              | N/A                      | 72.9          | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:

ABB AC800M High Integrity SIL3

SERH VERSION

PIU

SIL  
Cap

2013.1.02

-

N/A

Logic Solver Voting:

Architecture Type:

B

|                               |                  |                          |    |
|-------------------------------|------------------|--------------------------|----|
| HFT:                          | 1                | $\beta$ -factor [%]:     | 1  |
| MRT [Hours]:                  | 24               |                          |    |
| Proof Test Interval [Months]: | 12               | Proof Test Coverage [%]: | 99 |
| Proof Test Execution:         | Duration 1 hours |                          |    |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 68 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 11.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) Safety Instrumented Function is quantified as follows.

**Table 69 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 7.25E-03           | 2                              | N/A                      | 712.45        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1001 |
| $\beta$ -factor [%]:              | N/A  |

#### 11.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

|                      |      |                      |           |
|----------------------|------|----------------------|-----------|
| Voting within group: | 1001 | Voting type:         | Identical |
| HFT:                 | 0    | $\beta$ -factor [%]: | N/A       |

| Equipment Leg<br>(each): | SERH VERSION | 2 <sub>H</sub> | PIU | ACTYPE | SIL CAP |
|--------------------------|--------------|----------------|-----|--------|---------|
| Parker N                 |              | -              | -   | A      | 3       |
| Rotork                   |              | -              | -   | A      | 3       |



|                                             |           |   |   |   |   |
|---------------------------------------------|-----------|---|---|---|---|
| Pekos Floating Ball Valve, metal seat, < 8" | 2010.1.01 | - | - | A | 3 |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02 | - | - | B | 3 |

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

**Table 70 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |    |          |
|---------------------------------------------|----------------------|----------|----|----------|----|----|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU | NE       |
| Each Leg                                    |                      |          |    |          |    |    |          |
| Parker N                                    |                      | 5.70E-08 |    | 6.54E-07 |    |    |          |
| Rotork                                      |                      |          |    | 3.15E-08 |    |    |          |
| Pekos Floating Ball Valve, metal seat, < 8" |                      |          |    | 5.55E-07 |    |    | 2.02E-06 |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |    | 4.07E-08 |
| SFF [%]                                     |                      |          |    |          |    |    | 11.5     |
| ROUTE 2H COMPLIANT                          |                      |          |    |          |    |    | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 12 SIF-077 Temp efter O2-förångare vid tankparken (251TI300 |L2) => Close Avstängningsventil O2 efter tankpark (251SV520)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-077 Temp efter O2-förångare vid tankparken (251TI300 |L2) =>

Close Avstängningsventil O2 efter tankpark (251SV520) Safety Instrumented Function.

### 12.1 General SIF Information

The following characterizes the Safety Instrumented Function.

|                  |                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------|
| SIF Name:        | Temp efter O2-förångare vid tankparken (251TI300  L2) =><br>Close Avstängningsventil O2 efter tankpark (251SV520) |
| SIF Tag          | SIF-077                                                                                                           |
| SIF Description: | Temp efter O2-förångare vid tankparken (251TI300  L2) =><br>Close Avstängningsventil O2 efter tankpark (251SV520) |

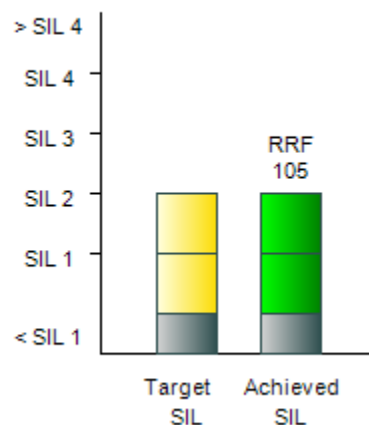
### 12.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF  $\geq 100$

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 105.5



### 12.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-077 Temp efter O2-förångare vid tankparken (251TI300 |L2) =>

Close Avstängningsventil O2 efter tankpark (251SV520) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

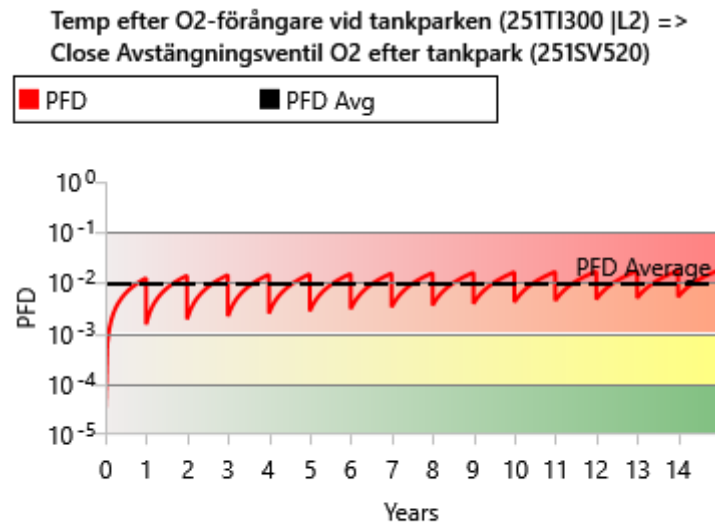
Mission Time: 15 years  
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:



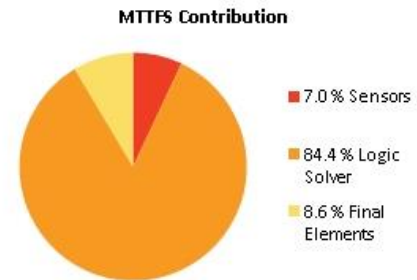
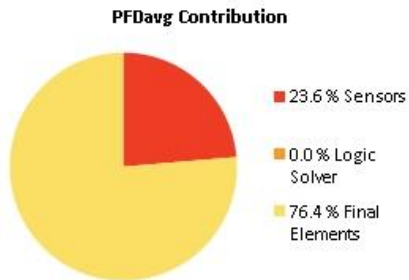
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-077 Temp efter O2-förångare vid tankparken (251TI300 |L2) =>

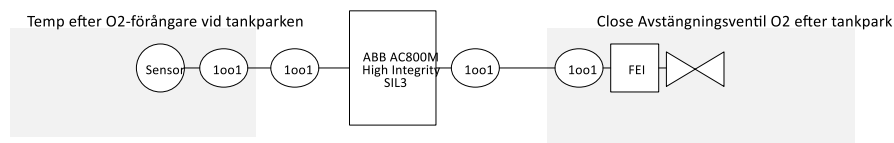
Close Avstängningsventil O2 efter tankpark (251SV520) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

**Table 71 Functional Safety Performance**

| PFD <sub>AVG</sub> | RRF   | ACHIEVED SIL       |                                |                          | MTTFS (YEARS) |
|--------------------|-------|--------------------|--------------------------------|--------------------------|---------------|
|                    |       | PFD <sub>AVG</sub> | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |
| 9.47E-03           | 105.5 | 2                  | 2                              | N/A                      | 61.17         |



The SIF Conceptual Design is modeled as shown in the picture below.



### 12.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-077 Temp efter O2-förångare vid tankparken (251TI300 | L2) =>

Close Avstängningsventil O2 efter tankpark (251SV520) Safety Instrumented Function is quantified as follows.

**Table 72 Sensor Part Functional Safety Performance**

| PFDavg   | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|----------|--------------------------------|--------------------------|---------------|-----|-----|
|          | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.24E-03 | 2                              | N/A                      | 873.51        | 0   | 2   |

Number of Sensor group(s): 1  
Voting between groups: 1oo1  
β-factor [%]: N/A

#### 12.3.1.1 Sensor Group 1: Temp efter O2-förångare vid tankparken

The information and reliability data underneath describe the Temp efter O2-förångare vid tankparken sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1      Voting type: Identical  
HFT: 0      β-factor [%]: N/A

Equipment Leg  
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

Endress+Hauser iTemp TMT 182                      2006.2.02                      -                      -                      B                      -

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]                      2013.1.02                      -                      -                      B                      3

MRT [Hours]:                      24

Proof Test Interval [Months]:                      12                      Proof Test Coverage [%]:                      90

Proof Test Execution:                      Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temp efter O2-förångare vid tankparken.

**Table 73 Reliability Data Sensor Group Temp efter O2-förångare vid tankparken**

| COMPONENT                                 | FAILURE RATES [1/HR]        |                             |                             |                             |                             |                             |                             |                             |                             |                             |
|-------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
|                                           | FAIL LOW                    | FAIL HIGH                   | FAIL DET.                   | SD                          | SU                          | DD                          | DU                          | AD                          | AU                          | NE                          |
| Each Leg                                  |                             |                             |                             |                             |                             |                             |                             |                             |                             |                             |
| Endress+Hauser iTemp TMT 182              | 1.66E-08<br><i>0.00E+00</i> | 9.16E-08<br><i>0.00E+00</i> | 9.18E-08<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>1.08E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 1.07E-07<br><i>1.99E-07</i> | 0.00E+00<br><i>0.00E+00</i> | 1.22E-09<br><i>1.22E-09</i> | 9.78E-08<br><i>9.78E-08</i> |
| ABB AC800M High Integrity SIL3 [AnalogIn] | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 4.28E-09<br><i>2.26E-08</i> | 4.28E-13<br><i>4.28E-13</i> | 1.83E-08<br><i>0.00E+00</i> | 1.83E-12<br><i>1.83E-12</i> | 0.00E+00<br><i>0.00E+00</i> | 0.00E+00<br><i>0.00E+00</i> | 1.09E-08<br><i>1.09E-08</i> |
|                                           |                             |                             |                             |                             |                             |                             | SFF [%]                     |                             | 40.0                        |                             |
|                                           |                             |                             |                             |                             |                             |                             | ROUTE 2H COMPLIANT          |                             | -                           |                             |

*The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.*

*The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.*

### 12.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-077 Temp efter O2-förångare vid tankparken (251TI300 | L2) =>

Close Avstängningsventil O2 efter tankpark (251SV520) Safety Instrumented Function is quantified as follows.

**Table 74 Logic Solver Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 2.01E-08           | 3                              | N/A                      | 72.46         | 1   | 2   |

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name:                      ABB AC800M High  
Integrity SIL3

|                                |              |     |         |
|--------------------------------|--------------|-----|---------|
| Equipment:                     | SERH VERSION | PIU | SIL Cap |
| ABB AC800M High Integrity SIL3 | 2013.1.02    | -   | N/A     |

|                               |                  |                          |    |
|-------------------------------|------------------|--------------------------|----|
| Logic Solver Voting:          |                  | Architecture Type:       | B  |
| HFT:                          | 1                | $\beta$ -factor [%]:     | 1  |
| MRT [Hours]:                  | 24               |                          |    |
| Proof Test Interval [Months]: | 12               | Proof Test Coverage [%]: | 99 |
| Proof Test Execution:         | Duration 1 hours |                          |    |

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

**Table 75 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3**

| COMPONENT                                            | FAILURE RATES [1/HR] |          |          |          |          |          |          |
|------------------------------------------------------|----------------------|----------|----------|----------|----------|----------|----------|
|                                                      | SD                   | SU       | DD       | DU       | AD       | AU       | NE       |
| Main Processor: PM865/TP830 SM811/TP868 (SM810/P855) | 4.51E-07             | 7.54E-07 | 1.46E-06 | 5.56E-11 | 3.63E-09 | 1.19E-08 | 6.79E-07 |
| Power Supply: 1*SD821 + 1*SS823 + 2*MCB              | 1.53E-06             | 9.06E-10 |          |          | 2.16E-08 | 4.52E-10 | 1.47E-07 |
| Digital Out Module: DO880/TU842/843                  | 3.98E-07             | 9.79E-09 | 4.17E-07 | 1.36E-10 | 3.89E-08 | 3.93E-10 | 3.68E-07 |
| Channel: Digital Out                                 | 8.53E-08             | 1.91E-08 |          |          |          |          | 4.07E-08 |
| Analog In Module: AI880A/TU844/845/TY801             | 3.69E-07             | 5.00E-10 | 3.84E-07 | 1.32E-10 | 3.89E-08 | 3.93E-10 | 2.11E-07 |
| Channel: Analog In                                   | 4.28E-09             | 4.28E-13 | 1.83E-08 | 1.83E-12 |          |          | 1.09E-08 |
|                                                      |                      |          |          |          |          | SFF [%]  | 100.0    |

### 12.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-077 Temp efter O2-förångare vid tankparken (251TI300 |L2) =>

Close Avstängningsventil O2 efter tankpark (251SV520) Safety Instrumented Function is quantified as follows.

**Table 76 Final Element Part Functional Safety Performance**

| PFD <sub>AVG</sub> | SIL LIMITS                     |                          | MTTFS (YEARS) | HFT | SSI |
|--------------------|--------------------------------|--------------------------|---------------|-----|-----|
|                    | ARCH. CONSTRAINTS<br>IEC 61511 | SYSTEMATIC<br>CAPABILITY |               |     |     |
| 7.25E-03           | 2                              | N/A                      | 712.45        | 0   | 2   |

|                                   |      |
|-----------------------------------|------|
| Number of Final Element group(s): | 1    |
| Voting between groups:            | 1001 |
| $\beta$ -factor [%]:              | N/A  |

#### 12.3.3.1 Final Element Group 1: Close Avstängningsventil O2 efter tankpark

The information and reliability data underneath describe the Close Avstängningsventil O2 efter tankpark final element group as it has been analyzed in this Safety Integrity Level verification.

|                      |      |                      |           |
|----------------------|------|----------------------|-----------|
| Voting within group: | 1001 | Voting type:         | Identical |
| HFT:                 | 0    | $\beta$ -factor [%]: | N/A       |

Equipment Leg  
(each):

|              |    |     |         |         |
|--------------|----|-----|---------|---------|
| SERH VERSION | 2H | PIU | AC TYPE | SIL CAP |
|--------------|----|-----|---------|---------|

|                                             |           |   |   |   |
|---------------------------------------------|-----------|---|---|---|
| Parker N                                    | -         | - | A | 3 |
| Rotork                                      | -         | - | A | 3 |
| Pekos Floating Ball Valve, metal seat, < 8" | 2010.1.01 | - | A | 3 |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 2013.1.02 | - | B | 3 |

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

|                               |                  |                          |    |
|-------------------------------|------------------|--------------------------|----|
| MRT [Hours]:                  | 24               |                          |    |
| Proof Test Interval [Months]: | 12               | Proof Test Coverage [%]: | 99 |
| Proof Test Execution:         | Duration 1 hours |                          |    |

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close Avstängningsventil O2 efter tankpark.

**Table 77 Reliability Data Final Element Group Close Avstängningsventil O2 efter tankpark**

| COMPONENT                                   | FAILURE RATES [1/HR] |          |    |          |    |                    |          |
|---------------------------------------------|----------------------|----------|----|----------|----|--------------------|----------|
|                                             | SD                   | SU       | DD | DU       | AD | AU                 | NE       |
| Each Leg                                    |                      |          |    |          |    |                    |          |
| Parker N                                    |                      | 5.70E-08 |    | 6.54E-07 |    |                    |          |
| Rotork                                      |                      |          |    | 3.15E-08 |    |                    |          |
| Pekos Floating Ball Valve, metal seat, < 8" |                      |          |    | 5.55E-07 |    |                    | 2.02E-06 |
| ABB AC800M High Integrity SIL3 [DigitalOut] | 8.53E-08             | 1.91E-08 |    |          |    |                    | 4.07E-08 |
|                                             |                      |          |    |          |    | SFF [%]            | 11.5     |
|                                             |                      |          |    |          |    | ROUTE 2h COMPLIANT | -        |

*The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis*

## 13 SILver™ Group Reuse Overview

This chapter provides an overview of the various groups that were defined during the SIL verification analyses of the Hybrit Pilot Plant project. The reuse of groups is available between Safety Instrumented Functions to account for identical hardware used in those different Safety Instrumented Functions. The group reuse allows users to specify a specific group once, reuse it, and with all future changes made to a single group cascading to all Safety Instrumented Functions it is used in. It is instrumental for the user to review all SIFs a single group is used in when changes are made to that group.

This overview indicates the groups that were reused, the number of times they were reused, and the Safety Instrumented Functions that they were used in.

### 13.1 Sensor Groups Reused

|                                             |                                    |
|---------------------------------------------|------------------------------------|
| Gastrück toppgas<br>(used 4 times)          | SIF-013, SIF-014, SIF-011, SIF-012 |
| Gastrück transferline<br>(used 3 times)     | SIF-021, SIF-024, SIF-020          |
| Tryckvakt O2-inj<br>(used 3 times)          | SIF-022, SIF-023, SIF-020          |
| Temperatur konvektionsdel<br>(used 2 times) | SIF-027, SIF-028                   |

### 13.2 Logic Solver Groups Reused

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABB AC800M High Integrity SIL3<br>(used 72 times) | SIF-996, SIF-997, SIF-998, SIF-026, SIF-001, SIF-034, SIF-002, SIF-003, SIF-004, SIF-005, SIF-051, SIF-052, SIF-053, SIF-054, SIF-056, SIF-058, SIF-059, SIF-060, SIF-061, SIF-062, SIF-063, SIF-076, SIF-077, SIF-064, SIF-065, SIF-067, SIF-068, SIF-069, SIF-070, SIF-071, SIF-074, SIF-073, SIF-072, SIF-075, SIF-019, SIF-013, SIF-014, SIF-017, SIF-018, SIF-081, SIF-039, SIF-040, SIF-042, SIF-045, SIF-046, SIF-048, SIF-011, SIF-021, SIF-024, SIF-022, SIF-023, SIF-020, SIF-025, SIF-027, SIF-028, SIF-029, SIF-015, SIF-012, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-037, SIF-038, SIF-041, SIF-043, SIF-044, SIF-047, SIF-080, SIF-079, SIF-082 |
| ABB AC800M High Integrity SIL3<br>(used 9 times)  | SIF-006, SIF-007, SIF-008, SIF-009, SIF-049, SIF-050, SIF-055, SIF-057, SIF-066                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### 13.3 Final Element Groups Reused

|                                                                 |                                                                        |
|-----------------------------------------------------------------|------------------------------------------------------------------------|
| Avstängning O2<br>(used 6 times)                                | SIF-026, SIF-037, SIF-043, SIF-044, SIF-079, SIF-082                   |
| Trip red.gaskompressor<br>(used 6 times)                        | SIF-050, SIF-025, SIF-043, SIF-044, SIF-047, SIF-038                   |
| Trip proc.gaskompressor<br>(used 5 times)                       | SIF-055, SIF-042, SIF-045, SIF-046, SIF-048                            |
| Trip red.gaskompressor<br>(used 2 times)                        | SIF-049, SIF-041                                                       |
| Avstängningsventil 2 överhettad ånga till PGH<br>(used 8 times) | SIF-076, SIF-016, SIF-017, SIF-018, SIF-028, SIF-041, SIF-047, SIF-080 |
| H2 till brännare PGH<br>(used 2 times)                          | SIF-019, SIF-079                                                       |
| NG till brännare PGH<br>(used 2 times)                          | SIF-019, SIF-082                                                       |
| Close NG till kylkrets                                          | SIF-042, SIF-045, SIF-046, SIF-048                                     |



(used 4 times)

Avstängning O2 PT12

(used 18 times)

NG till brännare PGH PT12

(used 5 times)

Avstängningsventil 2 överhettad ånga  
till PGH PT 6

(used 5 times)

SIF-021, SIF-022, SIF-024, SIF-023, SIF-020, SIF-025, SIF-027, SIF-029,  
SIF-015, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-041, SIF-047,  
SIF-081, SIF-038

SIF-025, SIF-027, SIF-029, SIF-031, SIF-032

SIF-012, SIF-043, SIF-044, SIF-079, SIF-082

## 14 Abbreviations and Definitions

### 14.1 Abbreviations

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| HFT             | Hardware Fault Tolerance                            |
| IPL             | Independent Protection Layers                       |
| MoonN           | "M" out of "N"                                      |
| MTTFS           | Mean Time To Fail Spurious                          |
| MRT             | Mean Repair Time                                    |
| PFD             | Probability of Failure on Demand                    |
| PFDavg          | Average Probability of Failure on Demand            |
| PFH             | Probability of a Dangerous Failure per Hour         |
| PIU             | Proven In Use                                       |
| PTC             | Proof Test Coverage                                 |
| PTI             | Proof Test Interval                                 |
| RRF             | Risk Reduction Factor                               |
| SERH            | Safety Equipment Reliability Handbook               |
| SFF             | Safe Failure Fraction                               |
| SIF             | Safety Instrumented Function                        |
| SIL             | Safety Integrity Level                              |
| SIS             | Safety Instrumented System                          |
| SSI             | Site Safety Index                                   |
| SRS             | Safety Requirements Specification                   |
| $\beta$ -factor | Beta factor, indicating common cause susceptibility |
| SD              | Safe Detected                                       |
| SU              | Safe Undetected                                     |
| DD              | Dangerous Detected                                  |
| DU              | Dangerous Undetected                                |
| AD              | Annunciation Detected                               |
| AU              | Annunciation Undetected                             |
| NE              | No Effect                                           |

### 14.2 Definitions

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Architectural Constraints | Indication of maximum SIL achievable of a SIF's subsystems architecture based on minimum hardware fault tolerance requirements. This is determined solely on the basis of Type A or Type B device selection, redundancy (hardware fault tolerance), and the safe failure fraction (calculated or conservatively assumed if no data is provided). It does not pertain to Systematic Capability or certification levels. Architectural Constraint limits are determined based on IEC 61508 or IEC 61511 tables. |
| Mission Time              | This is the time period that the SIF is expected to be operational. Alternatively this is the time period between points in time where the SIF can be considered as new. Note that this is not the proof test interval because of imperfect proof tests considerations. The mission time could, for example, correspond to the major turnaround period of a unit.                                                                                                                                             |
| Startup Time              | This represents the time it will take to perform maintenance and repair on a unit after a spurious trip plus the time it will take to restart the unit. This is typically longer than just the average repair time.                                                                                                                                                                                                                                                                                           |

## Systematic Capability

Indication of the maximum SIL achievable based on the SIF's equipment items systematic integrity alone. This is determined on the basis of the IEC 61508 certification level of each equipment item and/or the systematic integrity claim made by the user for a specific equipment item as part of its Proven In use justification. This addresses the requirement in IEC 61511 that all equipment used in a SIF must either be assessed per IEC 61508 or proven in use.

## 15 Disclaimer, Assumptions, Equipment Data

### 15.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

### 15.2 Assumptions SILver™

#### *SIL Capability*

If the user disables the **Consider IEC SIL Capability** option, it is assumed that the justification of equipment used is performed outside of the SILver tool as in that case the results of the SIL verification only address the quantitative requirements of IEC 61511.

A list of all other assumptions on which the SILver™ tool is based can be found in the exSILentia® user guide.

### 15.3 Equipment Data

*exida* has compiled a proprietary equipment failure database. This database is a compilation of failure data collected from a variety of public and confidential sources and presents an industry average. The database is published as the "Safety Equipment Reliability Handbook" ISBN-13: 978-1-934977-15-6. The reliability data collection process as described in this book applies to the SILver equipment data collection process.

The user is responsible for determining the applicability of the failure data to any particular environment. The stress levels assumed to determine the equipment failure rate are average worst case for an industrial environment. Descriptions of the environmental profiles used are included in the Safety Equipment Reliability Handbook. Accurate plant specific data is preferable to general industry average data. Industrial plant sites with high levels of stress must use failure rate data that is adjusted to a higher value to account for the specific conditions of the plant.